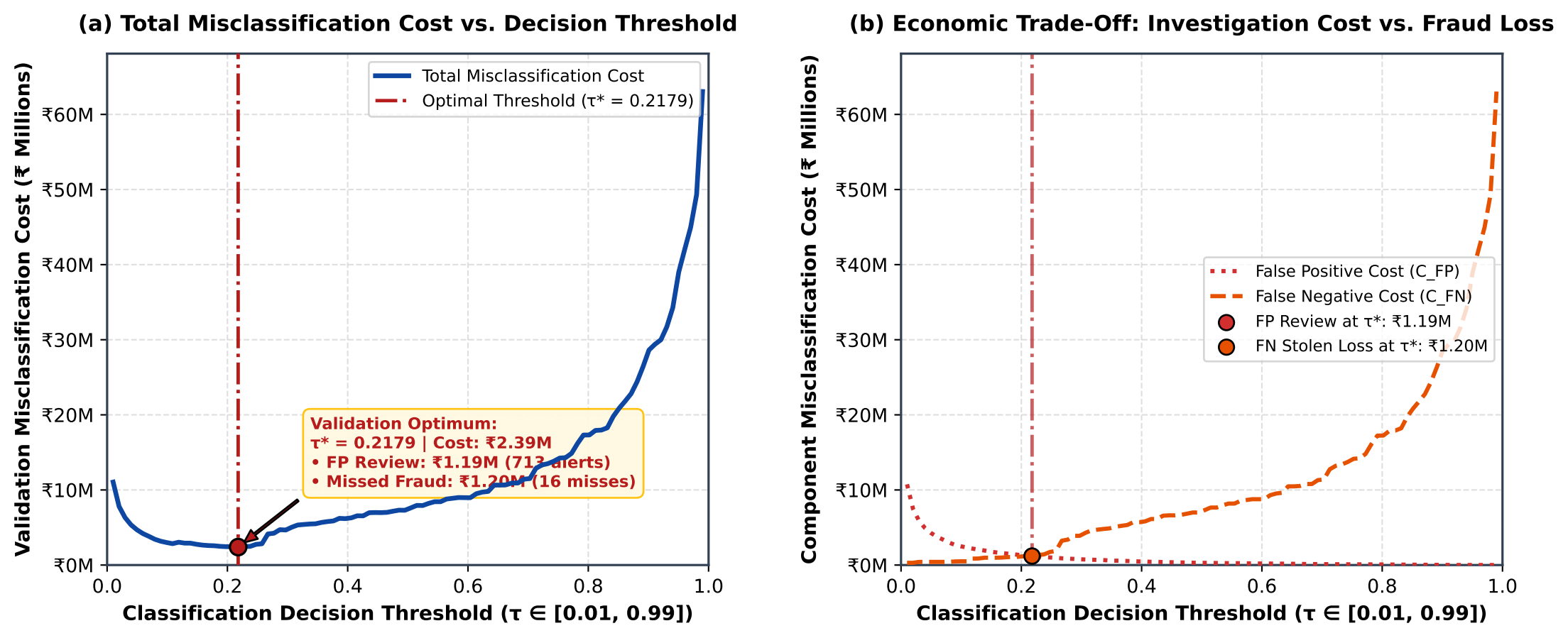
(a) Total Misclassification Cost vs. Decision Threshold
Total Misclassification Cost
Optimal Threshold (τ* = 0.2179)
Validation Optimum:
τ* = 0.2179 | Cost: ₹2.39M
• FP Review: ₹1.19M (713 alerts)
• Missed Fraud: ₹1.20M (16 misses)
Validation Misclassification Cost (₹ Millions)
₹0M
₹10M
₹20M
₹30M
₹40M
₹50M
₹60M
0.0
0.2
0.4
0.6
0.8
1.0
Classification Decision Threshold (τ ∈ [0.01, 0.99])
(b) Economic Trade-Off: Investigation Cost vs. Fraud Loss
False Positive Cost (C_FP)
False Negative Cost (C_FN)
FP Review at τ*: ₹1.19M
FN Stolen Loss at τ*: ₹1.20M
Component Misclassification Cost (₹ Millions)
₹0M
₹10M
₹20M
₹30M
₹40M
₹50M
₹60M
0.0
0.2
0.4
0.6
0.8
1.0
Classification Decision Threshold (τ ∈ [0.01, 0.99])